

Topology PhD Exam, May 2017

Work the following problems and show all work. Support all statements to the best of your ability.

1. Prove that a compact Hausdorff space is regular.
2. Prove that if $n > 0$, every map $S^n \rightarrow S^n$ is homotopic to a map with a fixed point.
3. The oriented surface M_g of genus g , embedded in $\mathbb{R}^3$ in the standard way, bounds a compact region R . Two copies of R , glued together by the identity map between their boundary surfaces M_g , form a closed 3-manifold X . Compute the homology groups of X using the Mayer–Vietoris sequence.
4. Let $f : S^1 \rightarrow \mathbb{R}$ be a continuous map. Prove that there exists a point $x \in S^1$ with $f(x) = f(-x)$. (Note: do not simply state the Borsuk–Ulam Theorem; give a direct proof.)
5. Let M be a closed simply-connected orientable 3-manifold. Compute the integral homology and cohomology of M . What can you say about $\pi_i(M)$, $i \leq 3$?

Answer the following with complete definitions, statements, or short proofs.

6. Prove that for a finite CW-complex X , $H^1(X; \mathbb{Z})$ is torsion-free.
7. Compute $\chi(\mathbb{CP}^3 \times \mathbb{RP}^2 \times S^2)$.
8. Give an example of a space that is path-connected but not locally path-connected.
9. State the Urysohn Lemma.
10. Prove that if $m \neq n$, then $\mathbb{R}^m$ is not homeomorphic to $\mathbb{R}^n$.
11. Does the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \arctan x$ admit a continuous extension $\bar{f} : \beta\mathbb{R} \rightarrow \mathbb{R}$ to the Stone–Čech compactification? What about the function $g(x) = e^x$?
12. State the Lefschetz Fixed Point Theorem.
13. Describe all the connected covering spaces $E \rightarrow S^1$.
14. Does the following exact sequence of abelian groups necessarily split? Prove or give a counterexample.
$$0 \rightarrow \mathbb{Z} \rightarrow A \rightarrow B \rightarrow 0$$
15. Compute the integral homology of the space $\mathbb{RP}^4 \times \mathbb{CP}^2$.